

The graph illustrates the computational complexity of three algorithms as a function of input size N . The x-axis, labeled 'Input size (N)', is on a logarithmic scale with major ticks at 100, 500, 1000, 5000, 10000, 50000, and 100000. The y-axis represents the number of iterations, with horizontal grid lines at intervals of 1000, ranging from 0 to 10000. Three data series are plotted: 'Naive' (red line with diamond markers), 'Gaussian elimination' (green line with star markers), and 'LU decomposition' (yellow line with square markers). The 'Naive' algorithm shows a significant increase in iterations as N grows, starting around 1000 for $N=100$ and reaching approximately 12000 for $N=100000$. In contrast, both 'Gaussian elimination' and 'LU decomposition' maintain a nearly constant number of iterations, staying below 1000 across the entire range of N .

Input size (N)	Naive (Iterations)	Gaussian elimination (Iterations)	LU decomposition (Iterations)
100	~1000	~500	~500
500	~1000	~500	~500
1000	~1000	~500	~500
5000	~1200	~500	~500
10000	~1300	~500	~500
50000	~8000	~500	~500
100000	~12000	~500	~500

